Skills Summary

Solving Systems Without a Graph

Use this reference while completing your worksheet or when studying.

Both methods do the same job: turn two equations in two unknowns into one equation in one unknown. They differ only in how they get rid of a variable — substitution replaces it, elimination cancels it. Always finish by checking the pair in *both* equations.

Skill 1 — Substitution

Use it when one equation is already solved for a variable, or is one step away. Replace that variable in the *other* equation, solve, then back-substitute for the second value.

Example: Solve $x + y = 15$ and $y = x + 3$.
Step 1. The second equation already gives y on its own, so it can be dropped straight into the first — no rearranging needed.
Step 2. $x + (x + 3) = 15$ gives $2x = 12$, so $x = 6$; then $y = 6 + 3 = 9$.
Check: $6 + 9 = 15$. **(6, 9)**

Try it: Solve $x + y = 20$ and $y = x + 4$.

check: **(8, 12)**

Skill 2 — Elimination

Use it when the same variable term appears in both equations. If the signs are opposite, add the equations; if they match, subtract. Either way one variable disappears.

Example: Solve $2x + y = 13$ and $x - y = 2$.
Step 1. The y terms are $+y$ and $-y$, opposite signs, so adding the two equations cancels y without any scaling.
Step 2. Adding gives $3x = 15$, so $x = 5$; then $5 - y = 2$ gives $y = 3$.
Check in the first: $2(5) + 3 = 13$. **(5, 3)**

Try it: Solve $3x + y = 14$ and $x - y = 2$.

check: **(4, 2)**

(!) Watch out: when you subtract, subtract *every* term, including the number on the right. Changing only the variable terms is the most common elimination slip.

Skill 3 — Choosing the method

Look before you solve: a variable already alone points to substitution; a matching term in both equations points to elimination; if neither is true, scale one equation until a term matches.

Example: Solve $4x + 3y = 27$ and $2x + 3y = 21$.
Step 1. Neither equation is solved for a variable, but both carry the identical term $3y$ — that points at elimination by subtraction.
Step 2. Subtracting gives $2x = 6$, so $x = 3$; then $4(3) + 3y = 27$ gives $3y = 15$ and $y = 5$.
Check in the second: $2(3) + 3(5) = 21$. **(3, 5)**

Try it: Solve $5x + 2y = 24$ and $3x + 2y = 16$.

check: **(4, 2)**